Quantum Field Theory IV Week 2 Homework

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + (D_\mu \phi)^\dagger (D^\mu \phi) - \lambda(\phi^\dagger \phi) + \mathcal{L}_{GF} \quad , \text{ where}$$

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2) \quad , \quad D_\mu = \partial_\mu + ieA_\mu \quad , \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$\mathcal{L}_{GF} = -\frac{1}{2\xi}(\partial_\mu A^\mu - \xi e \phi_1 \phi_2)^2$$

Using this version of covariant derivative following Peskin chapter 21.1, this allows us to cancel the mixing terms in quadratic part of the Lagrangian. Alternative to this would be to have $+\xi e \phi_1 \phi_2$ in $\mathcal{L}_{GF}$ inside the brackets.

## Exercise 2.8

We do a shift in $\phi_1$ : $\phi_1 \to \phi_1 + \phi_{cl}$ where $\phi_{cl}$ is constant valued classical field. $\Rightarrow \partial_\mu \phi_{cl} = 0$

$$\Rightarrow D_\mu \phi = (\partial_\mu \phi + ieA_\mu \phi) = \frac{1}{\sqrt{2}}\left[\partial_\mu(\phi_1 + i\phi_2) + ieA_\mu(\phi_1 + \phi_{cl} + i\phi_2)\right]$$

$$\Rightarrow D_\mu^\dagger \phi^\dagger = (\partial_\mu \phi^\dagger - ieA_\mu \phi^\dagger) = \frac{1}{\sqrt{2}}\left[\partial_\mu(\phi_1 - i\phi_2) - ieA_\mu(\phi_1 + \phi_{cl} - i\phi_2)\right]$$

$$\Rightarrow (D_\mu \phi)^\dagger (D^\mu \phi) = (D_\mu^\dagger \phi^\dagger)(D^\mu \phi) = \frac{1}{2}\left[(\partial_\mu \phi_1)^2 + (\partial_\mu \phi_2)^2 - 2e(\partial_\mu \phi_1)A^\mu \phi_2 + 2e(\partial_\mu \phi_2)A^\mu(\phi_1 + \phi_{cl}) + e^2 A^2\left[(\phi_1 + \phi_{cl})^2 + \phi_2^2\right]\right]$$

this works in this case

Here we can also write $A^2 = A_\mu g^{\mu\nu} A_\nu$

Now let's expand $\mathcal{L}_{GF}$ :

$$\mathcal{L}_{GF} = -\frac{1}{2\xi}\left[(\partial_\mu A^\mu)^2 - 2\xi e(\partial_\mu A^\mu)(\phi_1 + \phi_{cl})\phi_2 + \xi^2 e^2 (\phi_1 + \phi_{cl})^2 \phi_2^2\right]$$

$$\mathcal{L}_{GF} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2 + e(\partial_\mu A^\mu)\phi_{cl}\phi_2 + e(\partial_\mu A^\mu)\phi_1\phi_2 - \frac{\xi}{2}e^2\phi_{cl}^2\phi_2^2 - \frac{\xi}{2}e^2(\phi_1^2 + 2\phi_1\phi_{cl})\phi_2^2$$

$$\mathcal{L}_{GF} = \frac{1}{2\xi}A_\mu \partial^\mu \partial^\nu A_\nu - e\phi_{cl}A_\mu(\partial^\mu \phi_2) - \frac{\xi}{2}e^2\phi_{cl}^2\phi_2^2 + e(\partial_\mu A^\mu)\phi_1\phi_2 - \frac{\xi}{2}e^2(\phi_1^2 + 2\phi_1\phi_{cl})\phi_2^2$$

Here we used partial integration where we set the boundary integral to vanish. ($S = \int \mathcal{L} \Rightarrow$ partial integration). Now let's also expand $F_{\mu\nu}F^{\mu\nu} = (\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu)$

$$\Rightarrow F_{\mu\nu}F^{\mu\nu} = (\partial_\mu A_\nu)(\partial^\mu A^\nu) - (\partial_\mu A_\nu)(\partial^\nu A^\mu) - (\partial_\nu A_\mu)(\partial^\mu A^\nu) + (\partial_\nu A_\mu)(\partial^\nu A^\mu)$$

$$= 2(\partial_\mu A_\nu)(\partial^\mu A^\nu) - 2(\partial_\mu A_\nu)(\partial^\nu A^\mu)$$

// Let's again use partial integration techniques

$$= -2A_\nu \partial^2 A^\nu + 2A_\nu \partial_\mu \partial^\nu A^\mu$$

$$\Rightarrow -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}A_\mu \partial^2 g^{\mu\nu} A_\nu - \frac{1}{2}A_\mu \partial^\nu \partial^\mu A_\nu$$

$$= \frac{1}{2}A_\mu(\partial^2 g^{\mu\nu} - \partial^\nu \partial^\mu)A_\nu$$

Now let's write the fully expanded Lagrangian:

$$\mathcal{L} = \frac{1}{2} A_\mu \left( \partial^2 g^{\mu\nu} - \partial^\mu \partial^\nu + \frac{1}{\xi} \partial^\mu \partial^\nu + 2e^2 \phi_{cl}^2 g^{\mu\nu} \right) A_\nu + \frac{1}{2}(\partial_\mu \phi_1)^2 + \frac{1}{2}(\partial_\mu \phi_2)^2 - e(\partial_\mu \phi_1) A^\mu \phi_2$$

$$+ e(\partial_\mu \phi_2) A^\mu \phi_{cl} + e(\partial_\mu \phi_2) A^\mu \phi_1 + \frac{1}{2} e^2 A^2 \left[ \phi_1^2 + 2\phi_1 \phi_{cl} + \phi_2^2 \right] - e \phi_{cl} A^\mu (\partial_\mu \phi_2)$$

$$- \frac{\xi}{2} e^2 \phi_{cl}^2 \phi_2^2 + e(\partial_\mu A^\mu) \phi_1 \phi_2 - \xi/2 \; e^2 (\phi_1^2 + 2\phi_1 \phi_{cl}) \phi_2^2 - \lambda (\phi^\dagger \phi)^2$$

Now we expand $\lambda(\phi^\dagger \phi)^2$

$$\mathcal{L} = \frac{1}{2} A_\mu \left[ (\partial^2 + 2e^2 \phi_{cl}^2) g^{\mu\nu} - (1 - \tfrac{1}{\xi}) \partial^\mu \partial^\nu \right] A_\nu + \frac{1}{2}(\partial_\mu \phi_1)^2 + \frac{1}{2}(\partial_\mu \phi_2)^2 - \frac{1}{2}(\lambda \phi_{cl}^2 + \xi e^2 \phi_{cl}^2) \phi_2^2$$

$$- e(\partial_\mu \phi_1) A^\mu \phi_2 + e(\partial_\mu \phi_2) A^\mu \phi_1 + e^2 A^2 \phi_1 \phi_{cl} + e(\partial_\mu A^\mu) \phi_1 \phi_2 - \xi e^2 \phi_{cl} \phi_1 \phi_2^2 - 2\lambda \phi_{cl}^2 \phi_1^2$$

$$+ \frac{1}{2} e^2 A^2 \phi_1^2 + \frac{1}{2} e^2 A^2 \phi_2^2 - \frac{\xi}{2} e^2 \phi_1^2 \phi_2^2 - \frac{\lambda}{4} \left[ \phi_1^4 + \phi_2^4 + \phi_{cl}^4 \right] - \lambda \phi_1^3 \phi_{cl}$$

$$- \lambda \phi_{cl}^3 \phi_1 - \phi_{cl} \phi_1 \phi_2^2$$

In this exercise we are only interested in the quadratic part that is proportional to $A^2$ so to

$$\mathcal{L}_{A^2} = \frac{1}{2} A_\mu \left[ (\partial^2 + \underbrace{2e^2 \phi_{cl}^2}_{\text{Mass squared term for the A-field}, \; m_A^2 = 2e^2\phi_{cl}^2}) g^{\mu\nu} - (1 - \tfrac{1}{\xi}) \partial^\mu \partial^\nu \right] A_\nu$$

The part of the action for $A^2$-proportional cubic term is:

$$S_{A^2}[\phi_{cl}] = \frac{1}{2} \int d^4x \; A_\mu(x) \left[ (\partial^2 + 2e^2 \phi_{cl}^2) g^{\mu\nu} - (1 - \tfrac{1}{\xi}) \partial^\mu \partial^\nu \right] A_\nu(x)$$

$$= \frac{1}{2} \int d^4x \int d^4y \; A^\mu(x) \left[ (\partial^2 + 2e^2 \phi_{cl}^2) g_{\mu\nu} - (1 - \tfrac{1}{\xi}) \partial_\mu \partial_\nu \right] \delta(x-y) \, A^\nu(y)$$

$$S_{A^2}[\phi_{cl}] = \frac{1}{2} \int d^4x \int d^4y \; A^\mu(x) \, D^{-1}_{\mu\nu}(\phi_{cl}, x, y) \, A^\nu(y) \quad \text{, where}$$

$$D^{-1}_{\mu\nu}(\phi_{cl}, x, y) = \left[ (\partial^2 + 2e^2 \phi_{cl}^2) g_{\mu\nu} - (1 - \tfrac{1}{\xi}) \partial_\mu \partial_\nu \right] \delta(x-y)$$

The derminat we want to calculate is of form $\det\left[ - \frac{\delta^2 \mathcal{L}[\phi_{cl}]}{\delta K(x) \, \delta L(x)} \right]$ with $K, L \in \{\phi_1, \phi_2, A_\mu\}$ so we use the following $D$:

$$S_{A^2}[\phi_{cl}] = \frac{1}{2} \int d^4x \; A^\mu(x) \, D^{-1}_{\mu\nu}(\phi_{cl}, x) \, A^\nu(x) \quad \text{, with}$$

$$D^{-1}_{\mu\nu} = (\partial^2 + 2e^2 \phi_{cl}^2) g_{\mu\nu} - (1 - \tfrac{1}{\xi}) \partial_\mu \partial_\nu$$

Now let's write this in momentum space where $\partial_\mu \to i k_\mu \Rightarrow \partial^2 = -k^2$ since the fourie transform gives $\int \frac{d^4x}{(2\pi)^4} \partial_\mu e^{ik^\nu x_\nu} = \int \frac{d^4x}{(2\pi)^4} i k_\nu e^{ik^\nu x_\nu}$.

So in the momentum space we have

$$D_{\mu\nu}^{-1} = \left(-k^2 + 2e^2\phi_{cl}^2\right) g_{\mu\nu} + \left(1-\frac{1}{\xi}\right) k_\mu k_\nu$$

$$D^{-1}(\phi_{cl}, k)^\mu{}_\nu = \left[-\left(k^2 - 2e^2\phi_{cl}^2\right) g^\mu{}_\nu + \left(1-\frac{1}{\xi}\right) k^\mu k_\nu\right]$$

Now Let's choose a very convinient gauge, that is the Feynman gauge, so $\xi = 1$, this simplifies our $D^{-1}{}^\mu{}_\nu$ to

$$D^{-1}(\phi_{cl}, k)^\mu{}_\nu = -\left(k^2 - 2e^2\phi_{cl}^2\right) g^\mu{}_\nu$$

Now the remaining task is to compute the determinant:

$$\det\left(D^{-1}(\phi_{cl}, k)\right) \equiv \det\left(D^{-1}\right)^\mu{}_\nu$$

this is already diagonal

$\det(AB) = \det(A)\det(B)$

Now use $\det B = \exp\left[\mathrm{Tr}\left(\ln(B)\right)\right]$ from Peskin (9.77 For functional determinant

$$\det\left(D^{-1}(\phi_{cl}, k)\right) = \det\left(-\left(k^2 - 2e^2\phi_{cl}^2\right) g^\mu{}_\nu\right)$$

$$= \det(-g^\mu{}_\nu)\det\left(k^2 - 2e^2\phi_{cl}^2\right) = +\det\left(k^2 - 2e^2\phi_{cl}^2\right)$$

$$\det\left(D^{-1}(\phi_{cl}, k)\right) = +\exp\left[(VT)\int \frac{d^Dk}{(2\pi)^D} \ln\left[-k^2 - 2e^2\phi_{cl}^2\right]\right]$$

// $D = 4$ for this problem (in $\frac{d^Dk}{(2\pi)^D}$)

$$\det\left(D^{-1}(\phi_{cl}, k)\right) = +\exp\left[(VT)\int \frac{d^4k}{(2\pi)^4} \ln\left(k^2 - 2e^2\phi_{cl}^2\right)\right]$$

// VT is the volume term

Here we have used $\det(-1\cdot A) = (-1)^n \det(A)$ for $n\times n$ - matrix so $\det(-g^\mu{}_\nu) = (-1)^4 \det(g^\mu{}_\nu)$

We also used $\det(g^\mu{}_\nu) = \det(g^{\mu\alpha} g_{\alpha\nu}) = \det(g^{\mu\alpha})\det(g_{\alpha\nu}) = (-1)\cdot(-1)$

Another way to interpret this would be to think of $g^\mu{}_\nu$ as just the identity matrix or the kronecker delta $\delta^\mu{}_\nu$.

Exercise 2.9. The one-loop correction of $\Gamma_{eff}[\phi_a]$ is

$$\Delta\Gamma^{(1)}_{eff} \sim \frac{i}{2} \ln\det\left(-\frac{\delta\mathcal{L}}{\delta K \delta L}\right)\Big|_0$$

(I'm using peskin notation of $\frac{\delta\mathcal{L}}{\delta K \delta L}$ and not $\frac{\delta S}{\delta K \delta L}$)

where $K, L \in \{\phi_1, \phi_2, A_\mu\}$

$$\Rightarrow \Delta\Gamma^{(1)}_{eff} = \frac{i}{2}\Big[\underbrace{\ln\det\left(\phi_1^2\text{-contribution}\right) + \ln\det\left(\phi_2^2\text{-contribution}\right)}_{=\ O(2)\text{ - linear sigma model for which we got result last week}} + \underbrace{\ln\det\left(A^2\text{-contribution}\right)}_{\text{Calculated in previous exercise}}\Big]$$

First let's just list the terms of lagrangian that are quadratic in fields $(\phi_1, \phi_2)$

$$\Rightarrow \mathcal{L}_Q = \frac{1}{2}(\partial_\mu \phi_1)^2 + \frac{1}{2}(\partial_\mu \phi_2)^2 - \frac{1}{2}(\lambda \phi_{cl}^2 + \xi e^2 \phi_{cl}^2)\phi_2^2 - 2\lambda \phi_{cl}^2 \phi_1^2$$

We already chose the gauge $\xi = 1$ previously

$$= \frac{1}{2}(\partial_\mu \phi_1)^2 + \frac{1}{2}(\partial_\mu \phi_2)^2 - \frac{1}{2}(\lambda + e^2)\phi_{cl}^2 \phi_2^2 - \frac{1}{2}(4\lambda)\phi_{cl}^2 \phi_1^2$$

$$\mathcal{L}_Q = \frac{1}{2}(\partial_\mu \phi_1)^2 + \frac{1}{2}(\partial_\mu \phi_2)^2 - \frac{1}{2} m_2^2 \phi_2^2 - \frac{1}{2} m_1^2 \phi_1^2 \quad \text{, where } \begin{cases} m_1^2 = 4\lambda \phi_{cl}^2 \\ m_2^2 = (\lambda + e^2)\phi_{cl}^2 \end{cases}$$

Now we get the 1-loop correction of $V_{eff}$ by $\phi_1$ and $\phi_2$ using results from Peskin (11.78) and (11.79) (IN $\overline{MS}$ - scheme)

$$\Delta V_{\phi_1\phi_2} = \left[\frac{i}{2}\left(\ln \det(\phi_1^2\text{- contribution}) + \ln\det(\phi_2^2\text{- contribution})\right)\right] \cdot \left(-\frac{1}{VT}\right)$$

$$\Delta V_{\phi_1\phi_2} = -\frac{m_1^4}{4(4\pi)^2}\left(-\ln\left(\frac{m_1^2}{M^2}\right) + 3/2\right) - \frac{m_2^4}{4(4\pi)^2}\left(-\ln\left(\frac{m_2^2}{M^2}\right) + 3/2\right)$$

$$\Delta V_{\phi_1\phi_2} = +\frac{16\lambda^2 \phi_{cl}^4}{4(4\pi)^2}\left(\ln\left[\frac{4\lambda \phi_{cl}^2}{M^2}\right] - 3/2\right) + \frac{\phi_{cl}^4(\lambda+e^2)^2}{4(4\pi)^2}\left(+\ln\left(\frac{(\lambda+e^2)\phi_{cl}^2}{M^2}\right) - 3/2\right)$$

the extra $e^2$ term comes from my gauge choice of $\xi = 1$

Now let's calculate the 1-loop contribution from $A^2$ - terms:

$$\Delta \Gamma_{A^2}^{(1)} = \frac{i}{2} \ln \det(D^{-1}(\phi_{cl}, k)) = \frac{i}{2}(VT)\int \frac{d^4k}{(2\pi)^4} \ln(k^2 - 2e^2\phi_{cl}^2)$$

$$\Rightarrow \Delta V_{A^2} = -\frac{1}{VT}\Gamma_{A^2}^{(1)} = -\frac{i}{2}\int \frac{d^4k}{(2\pi)^4}\ln(k^2 - 2e^2\phi_{cl}^2)$$

$$\Rightarrow \Delta V_{A^2}^{(1)} = -\frac{i}{2}\int \frac{d^4k}{(2\pi)^4}\ln\left(\frac{k^2 - 2e^2\phi_{cl}^2}{k^2}\right)$$

We want to normalize this to be 1, when $\phi_{cl} = 0$ so that there is no contribution at no loop order so we add normalization of $\frac{1}{k^2}$ inside log

Now let's compute this using dimensional regularization:

$$\Delta V_{A^2}^{(1)} = -\frac{i}{2}\int \frac{d^dk}{(2\pi)^d}\partial_\alpha\left(\frac{1}{(k^2 - 2e^2\phi_{cl}^2)^\alpha} - \frac{1}{(k^2)^\alpha}\right)\Bigg|_{\alpha=0}$$

$$= +\frac{i}{2}\partial_\alpha \int \frac{d^dk}{(2\pi)^d}\frac{1}{(k^2 - \underbrace{2e^2\phi_{cl}^2}_{\equiv \Delta})^\alpha} - \frac{1}{(k^2)^\alpha}\Bigg|_{\alpha=0}$$

so that we have $(k^2-\Delta)^n$

we used $\ln\left(\frac{k^2 - 2e^2\phi_{cl}^2}{k^2}\right) = \ln(k^2 - 2e^2\phi_{cl}^2) - \ln(k^2)$ the $\ln(x) = -\partial_\alpha \frac{1}{x^\alpha}\Big|_{\alpha=0}$

Let's use the known results of these integral from Peskin eq. (A.44)

$$\Delta V_{A^2}^{(1)} = +\frac{i}{2}\partial_\alpha\left[\frac{(-1)^\alpha i}{(4\pi)^{d/2}}\frac{\Gamma(\alpha - d/2)}{\Gamma(\alpha)}\left(\frac{1}{\Delta}\right)^{\alpha - d/2} - \frac{(-1)^\alpha i}{(4\pi)^{d/2}}\frac{\Gamma(\alpha - d/2)}{\Gamma(\alpha)}(\rho)^{\frac{d}{2}-\alpha}\right]\Bigg|_{\alpha=0}$$

$\rho$: a small regulator

$$= +\frac{i}{2}\partial_\alpha\left[\underbrace{\frac{(1)\, i}{(4\pi)^{d/2}}\alpha\,\Gamma(-d/2)\left(\frac{1}{\Delta}\right)^{-d/2} - \frac{(1)\, i}{(4\pi)^{d/2}}\alpha\,\Gamma(-d/2)\,\rho^{d/2}}_{\text{near: } \alpha = 0 \,\Rightarrow\, \Gamma(\alpha) \approx \frac{1}{\alpha}}\right]\Bigg|_{\alpha=0}$$

$\Rightarrow$

$$\Delta V_A^{(1)} = +\frac{i}{2}\left[\frac{i}{(4\pi)^{d/2}}\,\Gamma(-d/2)\left(\frac{1}{\Delta}\right)^{-d/2} - \frac{i}{(4\pi)^{d/2}}\,\rho^{d/2}\,\Gamma(-d/2)\right]$$

$$= -\frac{1}{2}\frac{1}{(4\pi)^{d/2}}\,\Gamma(-d/2)\left(\Delta^{d/2} - \rho^{d/2}\right)$$

$$= -\frac{1}{2}\frac{1}{(4\pi)^{d/2}}\,\frac{\Gamma(2-d/2)}{\frac{d}{2}(d/2-1)}\left(\Delta^{d/2} - \rho^{d/2}\right)$$

Let's now expand $\Gamma(-d/2)$ using $\Gamma(n) = \frac{1}{n}\Gamma(n+1)$

$$\Rightarrow \Gamma(-d/2) = \frac{1}{-d/2}\cdot\Gamma(1-d/2)$$

$$= \frac{1}{-d/2}\frac{1}{(1-d/2)}\,\Gamma(2-d/2)$$

$$= \frac{1}{d/2(d/2-1)}\,\Gamma(2-d/2)$$

This is the pole we are interested in, the other poles at different values of d can be renormalized too, but we won't focus on that

Let's now take $\rho \to 0$ since it was just a small regulator also $\Delta V_A^{(1)} \to 0$ as $\Delta \to 0$ so there won't be any contribution to $V_{eff}$ at $n0$-loop order,

$$\Rightarrow \Delta V_A^{(1)} = -\frac{1}{2}\frac{1}{(4\pi)^{d/2}}\,\frac{\Gamma(2-d/2)}{\frac{d}{2}(d/2-1)}\,\Delta^{d/2}$$

Now we expand around $d = 4-\varepsilon$ :

$$\Delta V_A^{(1)} = \frac{1}{2}\frac{-1}{(4\pi)^2}\,(4\pi)^{\varepsilon/2}\,\frac{\Gamma(+\varepsilon/2)}{(2-\frac{\varepsilon}{2})(1-\varepsilon/2)}\,\Delta^{2-\varepsilon/2}$$

$$\frac{1}{(2-\varepsilon/2)(1-\varepsilon/2)} = \frac{1}{2-2\varepsilon-\varepsilon/2+\frac{\varepsilon^2}{4}}$$

$$\approx \frac{1}{2}\left(1+\frac{3}{4}\varepsilon - \frac{\varepsilon^2}{8}\right)$$

Binomial expansion

$$\Delta V_A^{(1)} = \frac{-1}{4}\frac{\Delta^2}{(4\pi)^2}\left[1+\frac{\varepsilon}{2}\ln(4\pi)\right]\left(1+\frac{3}{4}\varepsilon-\frac{\varepsilon^2}{8}\right)\left(\frac{2}{\varepsilon}-\gamma+O(\varepsilon)\right)\left(1-\frac{\varepsilon}{2}\ln(\Delta)\right)$$

$$\Delta V_A^{(1)} = -\frac{1}{4}\frac{\Delta^2}{(4\pi)^2}\left[\frac{2}{\varepsilon}-\gamma+\ln(4\pi)-\ln(\Delta)+\frac{3}{2}+O(\varepsilon)\right] \quad \| \varepsilon\to 0$$

$$(4\pi)^{\varepsilon/2} = \exp(\tfrac{\varepsilon}{2}\ln 4\pi) \approx 1+\frac{\varepsilon}{2}\ln(4\pi)$$

$$\Gamma(\varepsilon/2) = \left(\frac{2}{\varepsilon}-\gamma+O(\varepsilon)\right)$$

$$\Delta = m_A^2 = 2e^2\phi_{cl}^2$$

In $\overline{MS}$ we get

$$\Delta V_A^{(1)} = -\frac{1}{4}\frac{m_A^4}{(4\pi)^2}\left[-\ln\left(\frac{m_A^2}{M^2}\right)+\frac{3}{2}\right]$$ , where $M^2$ is a scale dependent quantity

$$\Delta V_A^{(1)} = +\frac{1}{4}\frac{4e^4\phi_{cl}^4}{(4\pi)^2}\left[\ln\left(\frac{2e^2\phi_{cl}^2}{M^2}\right)-\frac{3}{2}\right]$$

$$\ln\left(\frac{2e^2\phi_{cl}^2}{M^2}\right) = \ln\left(\frac{e^2\phi_{cl}^2}{M^2}\right)+\ln(2)$$

$$\Delta V_A^{(1)} = \frac{e^4\phi_{cl}^4}{(4\pi)^2}\left[\ln\left(\frac{e^2\phi_{cl}^2}{M^2}\right)+\ln(2)-3/2\right]$$

$\ln(2)-3/2 = -0.8068\ldots$

$-5/6 = -0.833\ldots$

So we just approximate $\ln(2)-3/2 \approx -5/6$

$$\Delta V_A^{(1)} \approx \frac{e^4\phi_{cl}^4}{(4\pi)^2}\left(\ln\left(\frac{e^2\phi_{cl}^2}{M^2}\right)-5/6\right)$$

The approximation $\ln(2)-3/2 = -5/6$ is the only way I can get $-5/6$ term so I've done that.

So the one loop correction from $A^2$ - terms is

$$\Delta V_{A^2}^{(1)} = \frac{e^4 \phi_{cl}^4}{(4\pi)^2} \left[ \ln\left( \frac{e^2 \phi_{cl}^2}{M^2}\right) - \frac{5}{6} \right]$$

and the total one loop correction to the effective potential is

$$\boxed{\begin{aligned} V_{eff}^{(1)}(\phi_{cl}) = {} & \frac{4\lambda^2 \phi_{cl}^4}{(4\pi)^2} \left[ \ln\left( \frac{4\lambda \phi_{cl}^2}{M^2} \right) - 3/2 \right] + \frac{(\lambda+e^2)^2 \phi_{cl}^4}{4(4\pi)^2} \left[ \ln\left( \frac{(\lambda+e^2)\phi_{cl}^2}{M^2} \right) - \frac{3}{2} \right] \\ & + \frac{e^4 \phi_{cl}^4}{(4\pi)^2} \left[ \ln\left( \frac{e^2 \phi_{cl}^2}{M^2} \right) - \frac{5}{6} \right] \end{aligned}}$$

<u>Exercise 2.10</u>

We have small couplings: $\lambda \sim e^2$

Now let's calculate the etremums of $V_{eff}^{(1)}(\phi_{cl})$

$$V_{eff}^{(1)}(\phi_{cl}) = \frac{\phi_{cl}^4}{(4\pi)^2} \left[ 4\lambda^2 \left( \ln\left( \frac{4\lambda\phi_{cl}^2}{M^2} \right) - 3/2 \right) + \frac{(\lambda+e^2)^2}{4} \left( \ln\left( \frac{(\lambda+e^2)\phi_{cl}^2}{M^2} \right) - 3/2 \right) + e^4 \left( \ln\left( \frac{e^2 \phi_{cl}^2}{M^2} \right) - 5/6 \right) \right]$$

$$\Rightarrow \frac{\partial V_{eff}^{(1)}(\phi_{cl})}{\partial \phi_{cl}} = 4 \frac{1}{\phi_{cl}} V_{eff}^{(1)}(\phi_{cl}) + \frac{\phi_{cl}^4}{(4\pi)^2} \left\{ \underbrace{\frac{\partial}{\partial \phi_{cl}} \left( 8\lambda^2 \ln(\phi_{cl}) + \frac{(\lambda+e^2)^2}{2} \ln(\phi_{cl}) + e^4 2 \ln(\phi_{cl}) \right)} \right\}$$

$$= \frac{4}{\phi_{cl}} V_{eff}^{(1)}(\phi_{cl}) + \frac{\phi_{cl}^4}{(4\pi)^2} \frac{1}{\phi_{cl}} \left( 8\lambda^2 + \frac{(\lambda+e^2)^2}{2} + e^2 2 \right)$$

Follows from basic properties of logarith where $\ln(ax) = \ln(a) + \ln(x)$
$\Rightarrow \partial_x \ln(ax) = \partial_x \ln(x) = \frac{1}{x}$

$$= \frac{4}{\phi_{cl}} V_{eff}^{(1)}(\phi_{cl}) + \frac{\phi_{cl}^3}{(4\pi)^2} \left[ 8\lambda^2 + 2e^2 + (\lambda+e^2)^2/2 \right]$$

The minimum is given by $\frac{\partial V_{eff}^{(1)}}{\partial \phi_{cl}} = 0$

$$\frac{\partial V_{eff}^{(1)}}{\partial \phi_{cl}} = \frac{4\phi_{cl}^3}{(4\pi)^2} \left[ 4\lambda^2 \left[ \ln\left( \frac{4\lambda\phi_{cl}^2}{M^2} \right) - 3/2 \right] + \frac{(\lambda+e^2)^2}{4} \left[ \ln\left( \frac{(\lambda+e^2)\phi_{cl}^2}{M^2} \right) - 3/2 \right] \right.$$
$$\left. + e^4 \left( \ln\left( \frac{e^2\phi_{cl}^2}{M^2} \right) - 5/6 \right) \right] + \frac{\phi_{cl}^3}{(4\pi)^2} \left( 4(4\lambda^2)\left(\frac{1}{2}\right) + 4\left(\frac{1}{2}e^2\right) + \frac{4}{4} \frac{(\lambda+e^2)^2}{2} \right)$$

$$0 = \frac{4\phi_{cl}^3}{(4\pi)^2} \left[ 4\lambda^2 \left( \ln\left( \frac{4\lambda\phi_{cl}^2}{M^2} \right) - 1 \right) + \frac{(\lambda+e^2)^2}{4} \left( \ln\left[ \frac{(\lambda+e^2)\phi_{cl}^2}{M^2} \right] - 1 \right) + e^2 \left( \ln\left( \frac{e^2\phi_{cl}^2}{M^2} \right) - \frac{5}{6} + \frac{1}{4} \right) \right]$$

This eq has solution at $\phi_{cl} = 0$ and at two another points $\phi_{cl} \neq 0$

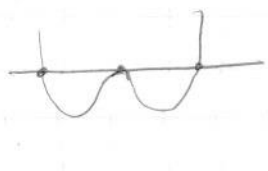
The ones at $\phi_{cl} \neq 0$ are given by

$$4\lambda^2 \left(\ln\left(\frac{4\lambda\phi_{cl}^2}{M^2}\right)-1\right) + \frac{(\lambda+e^2)^2}{4}\left(\ln\left(\frac{(\lambda+e^2)\phi_{cl}^2}{M^2}\right)-1\right) + e^2\left[\ln\left(\frac{e^2\phi_{cl}^2}{M^2}\right) - \frac{7}{12}\right] = 0$$

this equation has zeros at $\phi_{cl} \neq 0$ (there is actually pole at $\phi_{cl} = 0$
And these are actually minimums but it doesn't appear in $\frac{\partial V_{eff}}{\partial \phi_{cl}}$ since the
that are symmetric around y-axis. logarithms are multiplied by $\phi_{cl}^3$.)

We can look at simplified version of this by looking at example of function with similar form:

$$f(x) = a\,\phi^3 \left(b\left[\ln(c\,x^2) - 1\right]\right)$$ has the shape



with 3 extremums of which two are minimums with $x \neq 0$

Similarily $V_{eff}^{(1)}$ has local maximum at $\phi_{cl} = 0$ and minum at $\phi_{cl} \neq 0$ which implies spontaniously breaking symmetry

Since the couplings $\lambda$ and $e^2$ are small $(\lambda \sim e^2)$ the perturbations theory is valid since non of the terms in $V_{eff}^{(1)}$ behave badly at $\lambda \sim e^2 = $ small.

$\Rightarrow$ This effective potential leads to spontanious symmetry breaking.

Exercise 2.11

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - \mu^2 \phi^\dagger\phi - \lambda(\phi^\dagger\phi)^2 + \bar{\Psi}(i\not{\partial} - m)\Psi - \frac{y}{2}\phi\bar{\Psi}\Psi + h.c.$$

where $\phi$ is a complex scalar field and $\Psi$ is dirac fermion and $y$ is a real number

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2) \Rightarrow \phi^\dagger = \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2)$$

Let's compute the one loop effective potential by shifting the field $\phi_1$

$\phi_1(x) \to \phi_1(x) + \phi_{cl}$, where $\phi_{cl}$ is constant classical field

$$\mathcal{L} = (\partial_\mu\phi^\dagger)(\partial^\mu\phi) - \mu^2|\phi|^2 - \lambda(|\phi|^2)^2 + \bar{\Psi}(i\not{\partial}-m)\Psi - \frac{y}{2}\phi\bar{\Psi}\Psi - \frac{y}{2}\phi^\dagger\bar{\Psi}\Psi$$

Let's now expand the field $\phi$:

$$\Rightarrow |\phi|^2 = \frac{1}{2}\left[(\phi_1 + \phi_{cl})^2 + \phi_2^2\right], \quad \phi = \frac{1}{\sqrt{2}}(\phi_1 + \phi_{cl} + i\phi_2), \quad \phi^\dagger = (\phi_1 + \phi_{cl} - i\phi_2)\frac{1}{\sqrt{2}}$$

We don't need to worry about $(\partial_\mu\phi^\dagger)(\partial^\mu\phi)$ term since it won't affect the potential

$$\Rightarrow \mathcal{L} = (\partial_\mu \phi^\dagger)(\partial^\mu \phi) - \frac{\mu^2}{2}(\phi_1^2 + \phi_{cl}^2 + 2\phi_1\phi_{cl} + \phi_2^2) - \frac{\lambda}{4}(\phi_1^2 + \phi_{cl}^2 + 2\phi_1\phi_{cl} + \phi_2^2)^2$$

$$+ \bar\Psi(i\partial\!\!\!/ - m)\Psi - \frac{y}{2}(\phi_1 + \phi_{cl} + i\phi_2)\frac{1}{\sqrt2}\bar\Psi\Psi - \frac{y}{2}(\phi_1 + \phi_{cl} - i\phi_2)\frac{1}{\sqrt2}\bar\Psi\Psi$$

$$= (\partial_\mu\phi^\dagger)(\partial^\mu\phi) - \underline{\frac{\mu^2}{2}}\phi_1^2 - \frac{\mu^2}{2}\phi_{cl}^2 - \underline{\frac{\mu^2}{2}}\phi_2^2 - \mu^2\phi_1\phi_{cl} - \frac{\lambda}{4}[\phi_1^4 + \phi_{cl}^4 + \phi_2^4]$$

$$- \frac{\lambda}{4}[4\phi_1^3\phi_{cl} + \underline{6\phi_1^2\phi_{cl}^2} + 2\phi_1^2\phi_2^2 + 4\phi_1\phi_{cl}^3 + 4\phi_1\phi_{cl}\phi_2^2 + \underline{2\phi_{cl}^2\phi_2^2}]$$

$$+ \bar\Psi(i\partial\!\!\!/ - m - \frac{y\phi_{cl}}{\sqrt2})\Psi - y\phi_1\bar\Psi\Psi$$

The terms we are interested in are the $\left.\frac{\delta\mathcal{L}}{\delta K\,\delta L}\right|_0$ terms

for $K, L \in \{\phi_1, \phi_2, \Psi, \bar\Psi\}$

$$\Gamma[\phi_{cl}] = \left[\int d^4x\, \mathcal{L}_1[\phi_{cl}]\right] + \frac{i}{2}\ln\det\left(-\frac{\delta^2\mathcal{L}_1}{\delta\phi_1\delta\phi_1}\right) + \frac{i}{2}\ln\det\left(-\frac{\delta^2\mathcal{L}_1}{\delta\phi_2\delta\phi_2}\right) - i\ln\det\left(-\frac{\delta^2\mathcal{L}_1}{\delta\bar\Psi\delta\Psi}\right)$$
$$- i(\text{connected diagrams}) + \int d^4x\, \delta\mathcal{L}[\phi_{cl}]$$

the last term has $-i$ since for fermions (Grassman numbers): $\int D\bar\Psi D\Psi \exp(-\bar\Psi D\Psi) = \det(D) = \exp(\ln\det D)$

while the bosons have $\int D\phi \exp(-\phi_i B_{ij}\phi_j) = [\det(B)]^{-1/2} = \exp(-\frac{1}{2}\ln\det B)$

We will not need to write $\delta\mathcal{L}[\phi_{cl}]$ explicitely since we will use $\overline{MS}$ scheme

The determinants from the scalar $O(2)$- fields have been computed previously so we take the result from previous calculation/Peskin. Let's get on to calculating the determinant for $-\frac{\delta\mathcal{L}_1}{\delta\bar\Psi\delta\Psi}$

$$\ln\det\left(-\frac{\delta^2\mathcal{L}_1}{\delta\bar\Psi\delta\Psi}\right) = \ln\det\left[-\frac{\delta^2}{\delta\bar\Psi\delta\Psi}\left[\bar\Psi(i\partial\!\!\!/ - m - \frac{y\phi_{cl}}{\sqrt2})\Psi\right]\right]$$

$$= \ln\det\left[\frac{\delta}{\delta\Psi}(i\partial\!\!\!/ - m - \frac{y\phi_{cl}}{\sqrt2})\Psi\right] = \ln\det(i\partial\!\!\!/ - m - \frac{y\phi_{cl}}{\sqrt2})$$

$\frac{\delta^2}{\delta\bar\Psi\delta\Psi} = -\frac{\delta^2}{\delta\Psi\delta\bar\Psi}$ for Grassman numbers

$\frac{\delta\bar\Psi}{\delta\bar\Psi} = 1$

$$= \frac{1}{2}\left(\ln\det(i\partial\!\!\!/ - m - \frac{y\phi_{cl}}{\sqrt2}) + \ln\det(i\partial\!\!\!/ - m\,\frac{y\phi_{cl}}{\sqrt2})\right)$$

Now: $\ln\det D = \operatorname{tr}\ln D$

$$= \frac{1}{2}\left[\operatorname{tr}\ln(i\partial\!\!\!/ - m - \frac{y\phi_{cl}}{\sqrt2}) + \operatorname{tr}\ln(i\partial\!\!\!/ - m - \frac{y\phi_{cl}}{\sqrt2})\right]$$

Momentum space: $\partial\!\!\!/ \to i k\!\!\!/$

$$= \frac{1}{2}\left[\operatorname{tr}\ln(-k\!\!\!/ - m - \frac{y\phi_{cl}}{\sqrt2}) + \operatorname{tr}\ln(-k\!\!\!/ - m - \frac{y\phi_{cl}}{\sqrt2})\right]$$

$k\!\!\!/ = \gamma_\mu k^\mu$ ← gamma matrix

— Now we use $\operatorname{tr} \sim (VT)\int\frac{d^Dk}{(2\pi)^D}$

$$\Rightarrow \ln\det\left(-\frac{\delta^2\mathcal{L}_1}{\delta\bar{\psi}\,\delta\psi}\right) = \frac{1}{2}(VT)\left[\int\frac{d^Dk}{(2\pi)^D}\ \ln\left(-\gamma_\mu k^\mu - m - \frac{y\phi_{cl}}{\sqrt{2}}\right) + \ln\left(-\gamma_\mu k^\mu - m - \frac{y\phi_{cl}}{\sqrt{2}}\right)\right]$$

the integral is invariant under $k^\mu \to -k^\mu$ so we do just that for the second term

$$\Leftrightarrow \ln\det\left(-\frac{\delta^2\mathcal{L}_1}{\delta\bar{\psi}\,\delta\psi}\right) = \frac{1}{2}(VT)\left[\int\frac{d^Dk}{(2\pi)^D}\ \ln\left(-\not{k} - m - \frac{y\phi_{cl}}{\sqrt{2}}\right) + \ln\left(\not{k} - m - \frac{y\phi_{cl}}{\sqrt{2}}\right)\right]$$

Now we use power of Logarith: $\log(A) + \log(B) = \log(AB)$

$$\Rightarrow \ln\det\left(-\frac{\delta^2\mathcal{L}_1}{\delta\bar{\psi}\,\delta\psi}\right) = \frac{1}{2}(VT)\int\frac{d^Dk}{(2\pi)^D}\ \ln\left(-k^2 + \left(m + \frac{y\phi_{cl}}{\sqrt{2}}\right)^2\right)$$

|| Now use $\ln(x) = -\partial_\alpha \frac{1}{x^\alpha}\Big|_{\alpha=0}$

$$= -\frac{1}{2}(VT)\,\partial_\alpha\left[\int\frac{d^Dk}{(2\pi)^D}\ \frac{(-1)^\alpha}{\left(k^2 - \left(m + \frac{y\phi_{cl}}{\sqrt{2}}\right)^2\right)^\alpha}\right]\Bigg|_{\alpha=0}$$

|| write $m + \frac{y\phi_{cl}}{\sqrt{2}} = m_f$

$$= -\frac{1}{2}(VT)\,\partial_\alpha\left[\int\frac{d^Dk}{(2\pi)^D}\ \frac{(-1)^\alpha}{(k^2 - m_f^2)^\alpha}\right]\Bigg|_{\alpha=0}$$

|| Now we use Minkowski Space results from Peskin eq (A.44)

$$= -\frac{1}{2}(VT)\,\partial_\alpha\left[\frac{(-1)^\alpha(-1)^\alpha\, i}{(4\pi)^{d/2}}\ \frac{\Gamma(\alpha - d/2)}{\Gamma(\alpha)}\left(\frac{1}{m_f^2}\right)^{\alpha - d/2}\right]\Bigg|_{\alpha=0}$$

|| $\Gamma(\alpha) \to \frac{1}{\alpha}$ as $\alpha \to 0$ and $\partial_\alpha \alpha = 1$

$$= -\frac{1}{2}(VT)\ \frac{i}{(4\pi)^{d/2}}\ \frac{\Gamma(-d/2)}{(m_f^2)^{-d/2}}$$

|| Now let's expand with $d = 4 - \varepsilon$

$(4\pi)^{\varepsilon/2} = \exp\left(\frac{\varepsilon}{2}\ln(4\pi)\right) \simeq 1 + \frac{\varepsilon}{2}\ln(4\pi)$

$$= -\frac{i}{2}(VT)\ \frac{m_f^4}{(4\pi)^2}\left[\left(1 + \frac{\varepsilon}{2}\ln(4\pi)\right)\left(1 - \frac{\varepsilon}{2}\ln(m_f^2)\right)\Gamma(-d/2)\right]$$

$$= -\frac{i}{2}(VT)\ \frac{m_f^4}{(4\pi)^2}\left[\left(1 + \frac{\varepsilon}{2}\ln(4\pi) - \frac{\varepsilon}{2}\ln(m_f^2)\right)\left(1 + \frac{3}{4}\varepsilon\right)\left(\frac{2}{\varepsilon} - \gamma + \mathcal{O}(\varepsilon)\right)\right]$$

$$= -\frac{i}{4}(VT)\ \frac{m_f^4}{(4\pi)^2}\left[-\ln\left(\frac{m_f^2}{M^2}\right) + \frac{3}{2}\right]$$

|| here $M^2$ is scale dependent term

$\Gamma(-d/2) = \dfrac{\Gamma(2 - d/2)}{d/2\,(d/2 - 1)}$

$\dfrac{1}{d/2\,(d/2-1)} \simeq \dfrac{1}{2}\left(1 + \dfrac{3}{4}\varepsilon\right)$

↑ these approximations were done in more detail in previous calculation so leaving the detail out here.

$\Gamma(\varepsilon/2) = \frac{2}{\varepsilon} - \gamma + \mathcal{O}(\varepsilon)$

Where we've used the $\overline{MS}$-scheme

$$\Rightarrow \ln\det\left(-\frac{\delta^2\mathcal{L}_1}{\delta\bar{\psi}\,\delta\psi}\right) = -\frac{i}{4}(VT)\ \frac{m_f^4}{(4\pi)^2}\left[-\ln\left(\frac{m_f^2}{M^2}\right) + \frac{3}{2}\right]$$

Let's now also list the result of scalar particle contributed determinants:

The $\phi_1^2$ terms in Lagrangian are $-\frac{\mu^2}{2}\phi_1^2 - \frac{\lambda}{2}3\phi_{cl}^2\phi_1^2 \Rightarrow m_1^2 = \left(+\mu^2 + 3\lambda\phi_{cl}^2\right)$

And for $\phi_2^2$ : $-\frac{\mu^2}{2}\phi_2^2 - \frac{\lambda}{2}\phi_{cl}^2\phi_2^2 \Rightarrow m_2^2 = \left(\mu^2 + \lambda\phi_{cl}^2\right)$

or the eq (11.79) in Peskin and the section leading to it

Quoting the previous results, we know that the contribution to the effective potential from these is:

$$\Delta V^{(1)}_{eff\,\phi_1\phi_2} = \frac{1}{4}\frac{1}{(4\pi)^2}\left[ m_1^4\left[\ln\left(\frac{m_1^2}{M^2}\right) - \frac{3}{2}\right] + m_2^4\left[\ln\left(\frac{m_2^2}{M^2}\right) - \frac{3}{2}\right]\right]$$

, where $M^2$ is the scale dependent factor.

$$\Delta V^{(1)}_{eff\,\phi_1\phi_2} = \frac{1}{4}\frac{1}{(4\pi)^2}\left[ (\mu^2+3\lambda\phi_{cl}^2)^2\left[\ln\left(\frac{\mu^2+3\lambda\phi_{cl}^2}{M^2}\right) - \frac{3}{2}\right] + (\mu^2+\lambda\phi_{cl}^2)^2\left[\ln\left(\frac{\mu^2+\lambda\phi_{cl}^2}{M^2}\right) - \frac{3}{2}\right]\right]$$

And the contribution from $\ln\det\left(-\frac{\delta\mathcal{L}_1}{\delta\bar\psi\,\delta\psi}\right)$ is

$$\Delta V^{(1)}_{eff\,\bar\psi\psi} = -\frac{1}{(VT)}\left(-i\ln\left(\det\left(-\frac{\delta\mathcal{L}_1}{\delta\bar\psi\,\delta\psi}\right)\right)\right)$$

$$= -\frac{1}{VT}\left[-i\left(-\frac{i}{4}\right)(VT)\frac{m_f^4}{(4\pi)^2}\left(-\ln\left(\frac{m_f^2}{M^2}\right)+\frac{3}{2}\right)\right]$$

$$= \frac{m_f^4}{4(4\pi)^2}\left(-\ln\left(\frac{m_f^2}{M^2}+\frac{3}{2}\right)\right) \qquad \Bigg\| \quad m_f^2 = \left(m + \frac{y\phi_{cl}}{\sqrt{2}}\right)^2$$

$$= \frac{\left(m+\frac{y\phi_{cl}}{\sqrt2}\right)^4}{4(4\pi)^2}\left(-\ln\left[\frac{\left(m+\frac{y\phi_{cl}}{\sqrt2}\right)^2}{M^2}\right]+3/2\right)$$

It might or might not be usefull to have $M^2$ be different for this and the contributions from scalar-determinants but I'm leaving it as thy

Collecting these 1-loop contributions and the no-loop terms we get the One-loop effective potential of a Yukawa theory:

$$V_{eff}[\phi_a] = -\frac{\mu^2}{2}\phi_{cl}^2 - \frac{\lambda}{4}\phi_{cl}^4 + \frac{1}{4(4\pi)^2}\left[(\mu^2+3\lambda\phi_{cl}^2)^2\left(\ln\left[\frac{\mu^2+3\lambda\phi_{cl}}{M^2}\right]-\frac{3}{2}\right) + (\mu^2+\lambda\phi_{cl}^2)^2\left[\ln\left(\frac{\mu^2+\lambda\phi_{cl}^2}{M^2}\right)-\frac{3}{2}\right]\right.$$
$$\left. - \left(m+\frac{y\phi_{cl}^2}{\sqrt2}\right)^4\left[\ln\left(\frac{(m+y\phi_{cl})^2}{M^2}\right)-\frac{3}{2}\right]\right]$$

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Thanks for etending the deadline ☺. It took me some time to do and understand the stuff at 2.8 and 2.9 especially with the determinant and the Landau gauge. After a day of pondering and researching i found Peskin chapter 21.1 approach of gauge fixing to be easier which allowed me to compute determinant in Feynman gauge with diagonal elements, hopefully the approach wasn't too wrong. In problem 2.9 I had trouble reproducing $-5/6$ term in $V^{(1)}_{eff}$. I even tried different kinds of dimensional integrals and also different dimension but it was in vain so just decided to calculate normally and approximate $\ln(2) - 3/2 \doteq -5/6$.
Anyways thanks for the extension of deadline which allowed me to do problems 2.10 and 2.11 too !!